

SRIHARSHA N

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SUMMARY

Software, AI, and IoT developer with experience designing and developing integrated solutions across web applications, computer vision, cross-platform mobile development, and embedded systems. Proficient in Python, Django, Flask, and sensor integration, with a focus on translating complex requirements into practical, intelligent applications. Currently contributing to a robotic rover development project focused on lunar and Mars surface exploration, extending multidisciplinary expertise into planetary robotics and space technology.

PROJECT EXPERIENCE

1. WaveRoute AI — Intelligent Network Monitoring

- Developed a real-time dashboard that turns network data into clear, actionable insights.
- Mapped connected devices and monitored traffic to reveal network activity and router health.
- Added route analysis and explainable congestion assessment to help identify potential bottlenecks.

2. Smart Irrigation — IoT-Based Wireless Control

- Built an ESP32-based system to control irrigation remotely through wireless communication.
- Integrated sensors to monitor conditions and support watering decisions.
- Designed to simplify irrigation management and support efficient water use.

3. SafeHer — Smart Personal Safety Companion

- Developed a Flutter app that brings emergency support and trusted contacts into one accessible interface.
- Integrated GPS-based SOS alerts and direct emergency calls to help users seek assistance quickly.
- Added nearby safety facilities and multilingual support to make help easier to find.
- Included opt-in danger-sound detection, with a focus on privacy and user control..

Completed **15+ major projects** across software and hardware, translating engineering concepts into functional applications and integrated systems. Project experience spans artificial intelligence, computer vision, mobile applications, network monitoring, and IoT, grounded by a background in **Electronics and Instrumentation**.

Lunar & Mars Exploration Rover | Ongoing

Developing a robotic rover for planetary exploration, featuring subsurface ice detection for lunar water research and mineral sampling to investigate evidence of past water on Mars.

EDUCATION

M S RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE

OCT 2025 – AUG 2029

Bachelor of Engineering in Electronics and Instrumentation

- CGPA: 7.12

SRI CHAITANYA TECHNO SCHOOL

Central Board of Secondary Education

KEY SKILLS

Programming: Python, JavaScript, Dart, Embedded C++.

Web Development: HTML, CSS, Django, Flask, Tailwind CSS, Bootstrap, REST APIs.

AI & Computer Vision: OpenCV, YOLO, Machine Learning, Signal Processing.

Mobile Development: Flutter, Android Applications.

Embedded Systems & IoT: ESP32, Raspberry Pi, Sensor Integration, Wireless Communication, PCB Design, Circuit Schematic Design.

Databases & Tools: SQLite, MongoDB, Git, Chart.js, Vercel.